PRODUCT REQUIREMENTS DOCUMENT · V0.1 · STANDALONE PRODUCT

The lookahead that actually clears constraints.

A web product for superintendents on data center construction projects — one rolling plan with 3-week make-ready detail and a 6-week constraint and procurement view, starting in pre-commissioning.

This document defines what to build first, what success looks like, and what is deliberately out of scope. Every locked decision below comes directly from the founder's answers to the eight scoping questions. Where a decision was not made, it is listed as an open question in Section 14 — it is not assumed.

Status: Draft for review — v0.1	**Date:** November 8, 2026
Primary user: Superintendent (super)	**Product type:** Standalone web product — not a Safety Mojo module
Starting phase: Pre-commissioning	**Horizons:** 3-week detail + 6-week lookahead (one rolling plan)
Integrations: Phase 2 only	**Replaces today:** Project management tool, paper forms, whiteboards

Decisions locked in this PRD

Q1 — Today: PM tool / paper / whiteboard · Q2 — User: Supers · Q3 — Both horizons · Q4 — All failure modes · Q5 — Start: Pre-commissioning · Q6 — Integrations: Phase 2 · Q7 — All success metrics · Q8 — Standalone

> **The one-sentence thesis:** The lookahead already exists on almost every job. What fails is constraint clearance. This product wins not by drawing a prettier schedule, but by giving every constraint a named owner, a needed-by date, and escalating visibility until it is cleared — so the super stops firefighting problems that were visible three weeks earlier.

How to read this document

Sections 1–3 state the problem and who the product serves. Sections 4–9 are the buildable core: scope, horizons, data model, numbered functional requirements (FR-), workflows, and the commissioning model. Sections 10–14 cover metrics, non-goals, phasing, risks, and open questions. Appendix A is a worked example. Requirement IDs (FR-01, NFR-01) are stable — refer to them in design and engineering discussion.

1. The problem: prevention is invisible, firefighting is the job

On a data center job, the superintendent is the person who turns the plan into built work. In practice, much of that day is spent solving problems that were knowable weeks earlier: an unanswered request for information (RFI), a submittal nobody approved, gear that has not arrived, a predecessor trade that never committed, drawings that changed without reaching the field. Field leaders describe the ideal in exactly these terms — subs arriving knowing the work, material on site before it is needed, RFIs answered before the work is scheduled, and a week where nothing is on fire. The gap between that ideal and the daily reality is the product opportunity.

The founder confirmed that **all** of the candidate failure modes are in scope. That matters for design: this is not an RFI tracker or a delivery tracker. It is a constraint-clearance system that treats all five failure modes as one problem.

Failure mode (Q4: all)	What it looks like in the field	Cost when it surfaces late
Constraints never cleared	Constraint noted in a meeting, never owned, never chased	Crew stands idle; sequence collapses
RFIs / submittals stall	Answer arrives after the work was scheduled	Rework, out-of-sequence work, change-order disputes
Material / equipment misses	Long-lead item discovered short at install week	Idle crews; margin lost waiting
Trade commitments slip	Foreman agrees in the meeting, crew does not show or is undersized	Percent Plan Complete (PPC) drops; trust in the plan drops with it
Plan goes stale	Whiteboard or spreadsheet updated for the meeting, then ignored	Decisions made on last week's reality

Current state (Q1). Today the lookahead lives in a project management tool, on paper forms, or on whiteboards — often all three on the same job, none of them authoritative. Any replacement must therefore beat the whiteboard on speed: readable from across the trailer, printable for the wall, and updatable in the pull-planning meeting without a laptop ceremony. A product that is slower than the whiteboard will be abandoned by Friday.

2. Product principles

1. **Make-ready over reporting.** Every screen answers one question: what must happen, by whom, by when, so this work can start on time?
2. **Every constraint has an owner and a needed-by date.** No owner, no date — the system treats the constraint as unlogged and keeps surfacing it.
3. **Prevention gets credit.** Cleared-early constraints, kept commitments, and rising PPC are shown as wins. The planner who prevents the fire is made visible, not just the firefighter.
4. **Five minutes a day for the super.** Daily upkeep must fit between the morning huddle and the first walk. Bulk entry belongs in the weekly pull session, not in daily chores.
5. **One plan, two horizons.** The 3-week and 6-week views are filters on the same activities and constraints — never two documents that drift apart.

3. Users, buyers, and jobs to be done

Primary user (Q2): the superintendent. The super owns the lookahead, runs the weekly pull-planning session, chairs the daily huddle from the This Week view, and is accountable for PPC. The product is designed for the super first; every other role is a contributor or a viewer. If a design choice trades super speed against anyone else's convenience, the super wins in V1.

Role	Job to be done	Primary surface
Superintendent (primary)	Build and own the rolling plan; clear constraints; hold commitments	Full web app — trailer desktop + tablet
Trade foreman	Commit crew and dates; flag constraints early; confirm readiness	Lightweight commit / confirm view — mobile-friendly
Project manager / field engineer	Clear RFIs, submittals, procurement constraints they own	Constraint inbox filtered to their items
Commissioning authority (CxA) / Cx manager	Confirm level-by-level readiness; witness and sign gates	Commissioning readiness view (Section 9)
Owner / owner's rep (viewer)	See predictability: PPC trend, milestone confidence, top risks	Read-only dashboard — no editing

Buyer not yet confirmed — see Open Question OQ-1. The working assumption for V1 design only is that the general contractor (GC) buys per project. Nothing in V1 depends on the owner buying directly.

4. Scope: the pre-commissioning wedge

Starting phase (Q5): pre-commissioning. V1 covers the window from mechanical and electrical completion approaching, through pre-functional checks, energization, and readiness for functional performance testing. In commissioning terms that is the run-up through Level 2 and Level 3, with Level 4 and Level 5 milestones visible as targets (Section 9 defines the levels).

Why this wedge works: pre-commissioning concentrates every failure mode at once — punch items, vendor start-up, permits and authority-having-jurisdiction (AHJ) inspections, energization sequencing, test documentation — on a fixed, owner-visible deadline. Constraints here are concrete, named, and dated, which makes the product demonstrable in weeks rather than across a whole project lifecycle.

> **In V1:** one project, one or more buildings / data halls, pre-commissioning activities and their immediate predecessors. **Designed for, not built in V1:** earlier phases (structure, MEP rough-in, white-space fit-out) reuse the same objects and views and follow in V1.1 once the pre-commissioning workflow is proven (Section 12).

5. One rolling plan, two horizons

The product implements the Last Planner System pattern: a rolling lookahead sits between the master schedule and the weekly work plan. **Both horizons were selected (Q3)** — they are two lenses on the same data.

Lens	Horizon	Question it answers	Grain
Make-ready	3 weeks	Can this work actually start? What blocks it?	Activity broken to crew-level tasks; readiness per activity
Lookahead	6 weeks	What is coming, and what must be ordered, answered, or decided now?	Activities, deliveries, submittals, RFIs, permits, milestones
Filter	4 weeks	Middle view for GC reporting cadence	Same data, 4-week window — a saved filter, not a third plan
Weekly Work Plan	1 week	What do we commit to this week?	Committed tasks only; basis for PPC

Each week the window rolls forward one week: Week 6 enters, committed work drops into the Weekly Work Plan, and completed work feeds PPC and variance reasons. Activities carry planned start/finish, area or zone, system, responsible trade, crew size, and predecessors.

Figure 1 — The rolling window at a glance (illustrative)

Wk 1 — Commit	Wk 2	Wk 3	Wk 4	Wk 5	Wk 6 — Enters
Weekly Work Plan. Only READY work. PPC measured Friday.	Make-ready zone. Break to crew tasks, clear constraints, get commitments.		Lookahead zone. Spot deliveries, RFIs, submittals, permits and vendor bookings while there is still time to act.		

Reading it: work flows right to left as it approaches execution. A problem found in Week 5 is a planning task; the same problem found in Week 1 is a fire. The product's job is to move discovery from Week 1 to Weeks 4–6.

Board columns in the product (FR-01): Activity · Area / Zone · System · Trade · Crew size · Start / Finish · Predecessor · Readiness · Top constraint · Constraint owner · Needed-by. That column set is the direct digital successor to the whiteboard and paper form in use today (Q1) — familiar at a glance, with the owner and needed-by columns the whiteboard version usually loses.

6. Core objects and data model

Six objects carry the whole product. If a proposed feature does not create, change, or display one of these, it is probably out of scope for V1.

Object	Key fields
Activity	Name, area / zone, system, trade, planned start / finish, predecessor(s), crew size, readiness state, commissioning level (if any)
Constraint	Description, category, linked activity, **owner (named person)**, **needed-by date**, promised date, status, clearance date, escalation level
Area / System	Building, data hall / POD, room or zone; or system (e.g., UPS plant, CRAH loop, switchgear lineup) — activities and equipment attach to one or both
Equipment	Tag, description, vendor, required-on-site date, promised / actual delivery, start-up and witness needs — switchgear, UPS, generators, chillers / CRAH units and similar long-lead gear
Milestone	Name, target date, confidence, linked activities — e.g., energization, Level 3 complete, ready for Level 4, Integrated Systems Test (IST)
Commitment	Trade / foreman, task, promised date or week, kept / missed, variance reason when missed

Constraint categories (V1 fixed list): RFI, Submittal, Material / Equipment, Design / Drawing, Predecessor Work, Labor / Crew, Permit / Inspection (AHJ), Decision (owner / GC), Vendor / Start-up, Documentation / Test Records, Access / Space, Safety / Energization Permit. A fixed list keeps reporting comparable across weeks and projects; custom categories are a V1.1 candidate, not V1.

Readiness states — every activity in the 3-week window is in exactly one:

READY — all constraints cleared; can be committed. **CONSTRAINED** — one or more open constraints with a credible path before start. **AT RISK** — a constraint will miss its needed-by date, or has no owner / no date. Readiness is computed from linked constraints, never set by hand alone — a super override is allowed, is logged, and requires a note.

7. Functional requirements — plan and constraints

FR-01 Rolling board. Show activities in a week-column board across a selectable 3-, 4-, or 6-week window, grouped by area / system or by trade. Drag to reschedule within the window; changes are timestamped.

FR-02 Activity detail. One click from the board: dates, trade, crew, predecessors, linked constraints, equipment, documents, and readiness state with the reason for that state.

FR-03 Constraint log. A single sortable log of all constraints: linked activity, category, owner, needed-by, promised date, age, status (Open / Promised / Cleared / Missed). Filter by owner, category, week, and overdue.

FR-04 No orphan constraints. A constraint cannot be saved without an owner and a needed-by date. The UI blocks save and explains why, in one sentence.

FR-05 Escalation. Automatic escalation: owner notified at 7 days and 2 days before needed-by if still open; super notified on the needed-by date; project manager notified at 2 days overdue. Escalations are in-app and by email in V1.

7. Functional requirements (continued)

FR-06 RFI and submittal constraints. RFIs and submittals are tracked as constraint records in V1 (number, title, owner, needed-by, status) entered manually or by import. Live sync with the system of record is Phase 2 (Section 12).

FR-07 Equipment tracker. Long-lead and start-up-critical equipment list with required-on-site date vs. promised and actual dates; a slip automatically creates an AT RISK flag on linked activities.

FR-08 Weekly Work Plan and commitments. From the 3-week board, the super pulls READY (or explicitly overridden) activities into next week's plan. Each foreman confirms crew and dates — that confirmation is the commitment. Confirm by link, without a full account, is required so foremen actually use it.

FR-09 PPC and variance. At week end, each committed task is marked Done / Not Done. A missed task requires one variance reason from a fixed list: Predecessor, Material, Labor, Design / RFI, Inspection, Weather, Equipment, Planning Error, Other (with note). The product computes PPC = tasks completed / tasks committed, by week, project, trade, and area, with a rolling 6-week trend.

FR-10 Make the foreman's early question easy. Any foreman can raise a constraint from a task in two taps (what is missing, who can clear it, when it is needed). This directly answers the field question: how does the backing question surface before the wall closes — here, before the test, inspection, or energization is scheduled.

FR-11 Daily huddle view. A single-screen This Week view for the morning huddle: today's starts, today's constraints due, yesterday's misses, tomorrow's at-risk work. Designed to be read from 2 metres on a trailer display.

FR-12 Print for the wall. One-click printable 3-week board and constraint log (large type, no colour dependence) to replace the whiteboard and to carry into rooms without screens.

FR-13 Freshness guard. The plan shows its own staleness: last-updated timestamp, activities not touched in 7+ days highlighted, and a weekly roll-forward prompt every Monday. A stale plan must look stale.

FR-14 Roles and permissions. Super (edit all), PM / field engineer (edit own constraints, comment all), Foreman (commit, raise constraints), CxA (confirm gates), Owner viewer (read-only). Project-scoped access; no cross-project visibility by default.

FR-15 Import to start. CSV / spreadsheet import for activities, equipment, and constraints, plus a pre-commissioning template (Section 9 checklists) so a pilot job is loaded in under a day without any integration.

Non-functional requirements

NFR-01 Web app, responsive: trailer desktop first, tablet second, phone for foreman confirmations. **NFR-02** Board loads in under 2 seconds for a 500-activity project. **NFR-03** Full audit trail on dates, owners, and readiness overrides. **NFR-04** Offline behaviour is an open question (OQ-4) — V1 assumes trailer connectivity and must state that plainly to pilot users.

8. Key workflows

Weekly cycle (the heartbeat). (1) Monday — roll the window forward; review last week's PPC and variance reasons in 10 minutes. (2) Mid-week pull session (30–45 min) — trades review Weeks 2–3, raise constraints, make commitments for next week. (3) Daily — super works the huddle view; owners clear constraints from their inbox. (4) Friday — mark Done / Not Done; PPC posts automatically.

Constraint lifecycle. Raised → Owned (owner + needed-by confirmed) → Promised (owner gives a date) → Cleared (evidence or confirmation attached) or Missed (escalates, variance reason captured, linked activity re-flagged). Average days in each state is a product metric (Section 10), not just a project metric.

The question this workflow must surface early. In pre-commissioning the equivalent of the backing question is: Is the test procedure approved? Is the vendor booked? Is the energization permit signed? Is the AHJ inspection requested? Templates in FR-15 prompt for these per system so the question is asked by the system, not remembered by the busiest person on the job.

9. Commissioning model: L1–L5, energization, and IST

Every system in scope carries a level-by-level readiness gate. An activity cannot be marked READY for a level until that level's entry criteria are cleared as constraints. The levels follow standard data center commissioning practice:

Level	Meaning	What the lookahead must show
L1	Factory witness testing	Equipment tested at factory; reports received — a documentation constraint on delivery
L2	Installation and pre-functional checks	Equipment set, terminated, torqued, labelled; pre-functional checklists complete per system
L3	Start-up and functional checks	Vendor start-up booked and done; point-to-point and rotation checks; energization prerequisites
L4	Functional performance testing	System tested under load conditions against the sequence of operations; failures logged as constraints
L5	Integrated Systems Test (IST)	Whole-facility scenarios (e.g., utility loss) across systems; the terminal milestone for the phase

Energization is a first-class milestone, not a note. Each energization event has its own mini-gate: switchgear / UPS readiness, AHJ and utility approvals where applicable, lockout / tagout and energization permit, arc-flash labelling, and a named authoriser. No linked L3 activity can be READY before its energization gate clears.

Pre-commissioning templates (FR-15). Ship starter checklists per system type — UPS plant, switchgear lineup, generator plant, CRAH / chilled-water loop, fire alarm — each generating the standard activities and constraints for L2–L3 so a super starts from 80% complete, not a blank board.

10. Metrics and success criteria — all four (Q7)

The founder selected all four candidate measures. V1 must instrument all four from day one, even where the target is set after a baseline period on the pilot project.

Metric	Definition in product	How success is judged
Percent Plan Complete	Committed tasks completed / committed, weekly, with variance-reason breakdown	Rising trend over 8 weeks on the pilot; baseline set in Weeks 1–2, target agreed with the pilot GC — do not promise a number before baseline
Delay reduction	Activities starting on planned date; milestone forecast vs. target for energization and Ready for L4	Fewer late starts in the 3-week window vs. the project's prior-phase record
Constraint clearance	% constraints cleared by needed-by; median days from Raised to Cleared; % cleared early	Clear-by-needed-by rate improves month over month; zero constraints without owner / date
RFI cycle time	Days from RFI constraint raised to cleared, for RFI-category constraints	Median cycle time falls; RFIs answered before linked work is scheduled becomes the norm, tracked explicitly

Product-health metrics (internal): weekly active supers, roll-forward completion rate, foreman confirmation rate, and % of activities with a current readiness state. If supers stop rolling the plan forward, the product has failed regardless of schedule outcomes.

11. Non-goals for V1

- **Not a master scheduler.** No critical-path engine, no P6 replacement. The master schedule is an input reference in V1, a sync in Phase 2.
- **Not a general project management suite.** No document management, budgeting, daily logs, or punch-list system beyond constraints linked to lookahead work.
- **Not a Safety Mojo module (Q8).** Standalone product, standalone brand and data. Any future safety integration (e.g., pre-task plans) is a separate decision.
- **No integrations in V1 (Q6).** No P6, Procore, BIM, or accounting sync. Manual entry plus CSV import only — deliberately, to prove the workflow before the plumbing.
- **No AI scheduling in V1.** No auto-generated schedules or predicted dates. Readiness is rule-based from constraints so the super can trust and explain it.
- **Not multi-phase in V1.** Phases before pre-commissioning are V1.1 scope, using the same model.

12. Phasing

Phase	Scope
V1 — Pilot	FR-01 to FR-15, pre-commissioning only, one pilot project, CSV import, email notifications, print views
V1.1	Extend to MEP rough-in and white-space fit-out; custom constraint categories; portfolio view across projects for a GC
Phase 2 — Integrations	P6 / Microsoft Project schedule sync; Procore RFI, submittal and delivery sync; BIM / model area mapping; calendar and Slack / Teams notifications. Order set by pilot evidence, not in advance

13. Risks and assumptions

Risk / assumption	Mitigation in this PRD
Supers ignore another tool (highest risk)	Five-minute daily budget, print parity with the whiteboard, foreman confirm without an account, pilot with one committed super before scaling
Double entry vs. the PM tool kills trust	Be explicit: V1 is the field plan of record, the PM tool stays the commercial record; CSV import cuts retyping; integrations are Phase 2 by design
Foremen will not commit in writing	Commitment is one tap with crew size; PPC is reported by trade without public shaming defaults — the super controls sharing
Assumption: trailer connectivity	Stated in NFR-04; validate on the pilot site in Week 1 (OQ-4)
Assumption: GC is the buyer	Design does not depend on it; confirm in OQ-1 before pricing or contracting work begins

14. Open questions — decisions needed from the founder

OQ-1 Buyer and pricing. Who pays — GC per project, GC enterprise, or owner-mandated? Per project, per seat, or flat? Nothing in V1 pricing can be designed until this is answered.

OQ-2 Pilot. Which live project, which GC, which super, and what is the pre-commissioning start date we must be ready for? The pilot defines the template checklists that ship first.

OQ-3 Master schedule source. Even without integration, how will the pilot's activities arrive — P6 export, spreadsheet, or manual build from the template? Affects FR-15 import format priority.

OQ-4 Offline. Do pilot data halls have reliable connectivity? If not, offline tablet use moves from NFR note to V1 requirement and changes the build materially.

OQ-5 Naming and brand. Standalone product needs its own name and identity, separate from Safety Mojo. Needed before any external pilot conversation.

OQ-6 Success targets. After the 2-week baseline on the pilot, what PPC and clearance-rate targets will the GC sign up to? Targets are set with the pilot, not promised in sales material beforehand.

Recommended next step: Answer OQ-2 (pilot project and super) and OQ-1 (buyer) first. With a named pilot, the L2–L3 templates, import format, and baseline metrics all become concrete within one working session — without one, further PRD detail is speculation.

Appendix A — Worked example (illustrative)

The following is an illustrative example to make the model concrete. It is not a real project, customer, or schedule.

Setting: Data Hall 2, pre-commissioning. Milestone: Energize UPS Plant B, then Ready for Level 4 testing.

Example — how the 6-week and 3-week lenses show the same work

Week 6 lens: Generator load-bank test appears. Constraints raised now: load-bank rental (Material, owner: Field Engineer, needed-by: 3 weeks before test), fuel delivery, approved test procedure (Submittal).

Week 3 lens: UPS Plant B start-up is CONSTRAINED — vendor technician not yet booked (Vendor constraint, owner: PM, needed-by: this Friday). Energization gate shows AHJ inspection still Open, so L3 activities behind it cannot go READY.

Week 1 commitment: Electrical foreman commits 6-person crew to complete pre-functional checklists for CRAH units 1–8. Friday: 7 of 8 done — miss reason: Predecessor (one unit's piping flush incomplete). PPC for the week posts at 86% with that single variance explaining the gap.

Sample constraint log — same week, as the super sees it

Constraint	Category	Owner	Needed-by	Status
Vendor technician booked for UPS B start-up	Vendor	PM	This Friday	Open — escalates Thursday
AHJ inspection for energization	Permit / Inspection	Field Engineer	Next Tuesday	Promised
Load-bank rental confirmed	Material	Field Engineer	3 wks before test	Cleared early
Approved generator test procedure	Submittal	PM	2 wks before test	Open
Piping flush complete for CRAH 8	Predecessor	Mechanical foreman	Last Thursday	Missed — variance logged

Note the shape: five rows explain the entire week — one miss with a reason, one escalation in flight, one early clearance worth crediting, and two items owned and dated. That is the whole product in miniature.

Appendix B — Glossary

Term	Meaning in this product
Lookahead	The rolling multi-week window between the master schedule and the weekly work plan, used to make work ready
Make-ready	The process of clearing every constraint on an activity before its planned start
Constraint	Anything that would prevent an activity starting or finishing as planned, with an owner and needed-by date
Weekly Work Plan	The set of tasks trades commit to for the coming week; the denominator and numerator of PPC
PPC — Percent Plan Complete	Committed tasks completed divided by committed tasks, per week; the core reliability metric
Variance reason	The coded cause recorded when a committed task is not completed
Pull planning	Collaborative session where trades plan backwards from a milestone and make commitments to each other
Last Planner System	The production-control method (master plan, phase plan, lookahead, weekly work plan, PPC learning) this product implements
Cx / CxA	Commissioning / Commissioning Authority — the party verifying systems perform per the owner's requirements
L1–L5	Commissioning levels: factory testing, installation / pre-functional, start-up, functional performance, integrated systems testing
IST	Integrated Systems Test — Level 5 whole-facility scenario testing
RFI / Submittal	Request for Information / product or procedure submission for approval — both tracked as constraint categories in V1
AHJ	Authority Having Jurisdiction — the inspecting / permitting authority
POD / Data Hall	A discrete, repeatable unit of data center capacity; the primary area grouping on most projects
Super	Superintendent — the primary user of this product

Document end — PRD v0.1, November 8, 2026. Changes from review should be made against requirement IDs (FR- / NFR-) and open questions (OQ-) so v0.2 can show a clean decision log.

> **Assumptions made on the founder's behalf (flagged, not hidden):** (1) GC as the working buyer assumption for design only — OQ-1 remains open. (2) Trailer desktop first, phone for foreman confirmations — no surface was specified in Q1–Q8. (3) 4-week treated as a saved filter on the same plan, since Q3 selected both 3- and 6-week horizons. (4) Email plus in-app notifications in V1; Slack / Teams held to Phase 2 with other integrations.